

ASSIGNMENTS – OOADI

LECTURE 7

1. Code example
 - a. Run the race-condition program found on Moodle
 - b. Does it produce the desired output? Why not?
 - c. Discuss ways to fix the program and make it count correctly
 - d. Implement and test if your solution(s) works

2. (LEGO) train fiddling
 - a. Design and implement user input for a train management program as a thread (you can call it a Command Line Input Handler) and the part which holds the list of trains in another thread. Make the program work with those two threads. Hint: create a class specific for 1) the user input and 2) the train list management.
 - b. Create a set of threads of the user input handler to emulate a few train operators that wishes to seek, insert, delete trains from the main list. Make this run.
 - c. Now automatize the user input to emulate 10-100's of train operators trying to query the train list (by either searching, inserting, deleting, updating train entries). You should use the Random class to randomly select between the actions (previously taken as input from a real human via keyboard) a thread/train operator can take (if you forgot, then see previous assignment where you also used the Random class).
 - d. How many threads/train operators can you handle before you start seeing problems? Is it possible to solve the issues that you observe? And to what level can your solution hold, i.e. how many threads can you handle before more issues shows up?